Function

Mind map

Definition

- Each $x \in A$: exactly one image
- Vertical line test
- Range $\subseteq$ co-domain

Domain

- Denominator $\neq 0$, root ≥ 0
- log: arg > 0, base $> 0, \neq 1$
- Do not simplify first

Standard functions

- Trig, a^x, $\log_a x$
- Polynomial, rational
- $x \pm \frac{1}{x}$, x^3

Modulus, signum

- $|x| \in [0, \infty)$
- $\operatorname{sgn} x \in \{-1, 0, 1\}$
- $x \operatorname{sgn} x = |x|$

GIF, fractional part

- $[x] \leq x < [x] + 1$
- $\{x\} = x - [x] \in [0, 1)$
- $[f] = k \iff k \leq f < k + 1$

Range

- Solve $x = g(y)$
- Quadratic in x: $D \geq 0$
- Inner range, then outer

Identical functions

- Same domain
- Same values

Graph transformations

- Shifts: $f(x + a)$, $f(x) + b$
- $|f(x)|$, $f(|x|)$, $|y| = f(x)$
- $\frac{ax+b}{cx+d}$: hyperbola

One-one / many-one

- Horizontal line test
- f' keeps one sign
- Cubic: $b^2 \leq 3ac$

Onto / into

- Range = co-domain
- Odd degree polynomial: onto
- Bijective = invertible

Counting functions

- Total n^m
- One-one ${}^nP_m$
- Onto: inclusion-exclusion

Functional equations

- Put $g(x) = t$
- $x \to \frac{1}{x}$, $1 - x$: two equations
- $f(x + y) = f(x) + f(y) \Rightarrow kx$
- $f(x) + f(1 - x) = c$: pair terms

Composite

- $D = \{x : f(x) \in D_g\}$
- Piecewise: use graph of inner
- f^n: find the cycle

Even / odd

- $f(-x) = \pm f(x)$
- Even + odd split
- Derivative swaps parity

Inverse

- Needs bijection
- Swap x, y and solve
- Mirror in $y = x$

Periodic

- $f(ax + b)$: $\frac{T}{|a|}$
- LCM, then test half
- Rational with irrational: aperiodic